

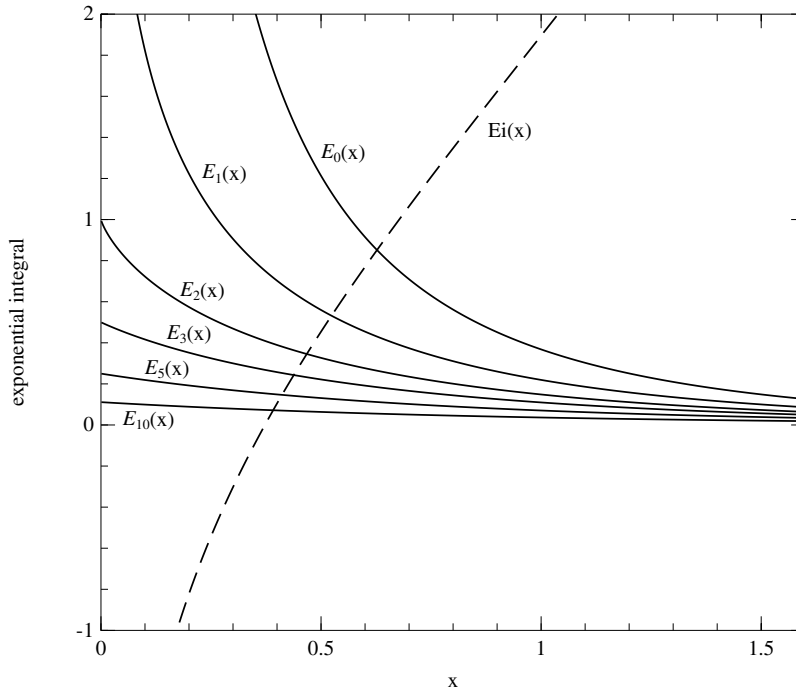


Figure 6.3.1. Exponential integrals $E_n(x)$ for $n = 0, 1, 2, 3, 5$, and 10 , and the exponential integral $Ei(x)$.

CITED REFERENCES AND FURTHER READING:

Abramowitz, M., and Stegun, I.A. 1964, *Handbook of Mathematical Functions* (Washington: National Bureau of Standards); reprinted 1968 (New York: Dover); online at <http://www.nist.gov/aands>, Chapters 6, 7, and 26.[1]

Pearson, K. (ed.) 1951, *Tables of the Incomplete Gamma Function* (Cambridge, UK: Cambridge University Press).

6.3 Exponential Integrals

The standard definition of the exponential integral is

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n} dt, \quad x > 0, \quad n = 0, 1, \dots \quad (6.3.1)$$

The function defined by the principal value of the integral

$$Ei(x) = - \int_{-x}^\infty \frac{e^{-t}}{t} dt = \int_{-\infty}^x \frac{e^t}{t} dt, \quad x > 0 \quad (6.3.2)$$

is also called an exponential integral. Note that $Ei(-x)$ is related to $-E_1(x)$ by analytic continuation. Figure 6.3.1 plots these functions for representative values of their parameters.

The function $E_n(x)$ is a special case of the incomplete gamma function

$$E_n(x) = x^{n-1} \Gamma(1-n, x) \quad (6.3.3)$$

We can therefore use a similar strategy for evaluating it. The continued fraction — just equation (6.2.6) rewritten — converges for all $x > 0$:

$$E_n(x) = e^{-x} \left(\frac{1}{x+} \frac{n}{1+} \frac{1}{x+} \frac{n+1}{1+} \frac{2}{x+} \dots \right) \quad (6.3.4)$$

We use it in its more rapidly converging even form,

$$E_n(x) = e^{-x} \left(\frac{1}{x+n-} \frac{1 \cdot n}{x+n+2-} \frac{2(n+1)}{x+n+4-} \dots \right) \quad (6.3.5)$$

The continued fraction only really converges fast enough to be useful for $x \gtrsim 1$. For $0 < x \lesssim 1$, we can use the series representation

$$E_n(x) = \frac{(-x)^{n-1}}{(n-1)!} [-\ln x + \psi(n)] - \sum_{\substack{m=0 \\ m \neq n-1}}^{\infty} \frac{(-x)^m}{(m-n+1)m!} \quad (6.3.6)$$

The quantity $\psi(n)$ here is the digamma function, given for integer arguments by

$$\psi(1) = -\gamma, \quad \psi(n) = -\gamma + \sum_{m=1}^{n-1} \frac{1}{m} \quad (6.3.7)$$

where $\gamma = 0.5772156649 \dots$ is Euler's constant. We evaluate the expression (6.3.6) in order of ascending powers of x :

$$\begin{aligned} E_n(x) = & - \left[\frac{1}{(1-n)} - \frac{x}{(2-n) \cdot 1} + \frac{x^2}{(3-n)(1 \cdot 2)} - \dots + \frac{(-x)^{n-2}}{(-1)(n-2)!} \right] \\ & + \frac{(-x)^{n-1}}{(n-1)!} [-\ln x + \psi(n)] - \left[\frac{(-x)^n}{1 \cdot n!} + \frac{(-x)^{n+1}}{2 \cdot (n+1)!} + \dots \right] \end{aligned} \quad (6.3.8)$$

The first square bracket is omitted when $n = 1$. This method of evaluation has the advantage that, for large n , the series converges before reaching the term containing $\psi(n)$. Accordingly, one needs an algorithm for evaluating $\psi(n)$ only for small n , $n \lesssim 20 - 40$. We use equation (6.3.7), although a table lookup would improve efficiency slightly.

Amos [1] presents a careful discussion of the truncation error in evaluating equation (6.3.8) and gives a fairly elaborate termination criterion. We have found that simply stopping when the last term added is smaller than the required tolerance works about as well.

Two special cases have to be handled separately:

$$\begin{aligned} E_0(x) &= \frac{e^{-x}}{x} \\ E_n(0) &= \frac{1}{n-1}, \quad n > 1 \end{aligned} \quad (6.3.9)$$

The routine `expint` allows fast evaluation of $E_n(x)$ to any accuracy EPS within the reach of your machine's precision for floating-point numbers. The only modification required for increased accuracy is to supply Euler's constant with enough significant digits. Wrench [2] can provide you with the first 328 digits if necessary!

`expint.h`

`Doub expint(const Int n, const Doub x)`

Evaluates the exponential integral $E_n(x)$.

```
{
    static const Int MAXIT=100;
    static const Doub EULER=0.577215664901533,
        EPS=numeric_limits<Doub>::epsilon(),
        BIG=numeric_limits<Doub>::max()*EPS;
    Here MAXIT is the maximum allowed number of iterations; EULER is Euler's constant
     $\gamma$ ; EPS is the desired relative error, not smaller than the machine precision; BIG is
    a number near the largest representable floating-point number.

    Int i,ii,nm1=n-1;
    Doub a,b,c,d,del,fact,h,psi,ans;
    if (n < 0 || x < 0.0 || (x==0.0 && (n==0 || n==1)))
        throw("bad arguments in expint");
    if (n == 0) ans=exp(-x)/x;                Special case.
    else {
        if (x == 0.0) ans=1.0/nm1;            Another special case.
        else {
            if (x > 1.0) {                    Lenz's algorithm (§5.2).
                b=x+n;
                c=BIG;
                d=1.0/b;
                h=d;
                for (i=1;i<=MAXIT;i++) {
                    a = -i*(nm1+i);
                    b += 2.0;
                    d=1.0/(a*d+b);            Denominators cannot be zero.
                    c=b+a/c;
                    del=c*d;
                    h *= del;
                    if (abs(del-1.0) <= EPS) {
                        ans=h*exp(-x);
                        return ans;
                    }
                }
                throw("continued fraction failed in expint");
            } else {
                ans = (nm1!=0 ? 1.0/nm1 : -log(x)-EULER);    Evaluate series.
                fact=1.0;                                     Set first term.
                for (i=1;i<=MAXIT;i++) {
                    fact *= -x/i;
                    if (i != nm1) del = -fact/(i-nm1);
                    else {
                        psi = -EULER;                Compute  $\psi(n)$ .
                        for (ii=1;ii<=nm1;ii++) psi += 1.0/ii;
                        del=fact*(-log(x)+psi);
                    }
                    ans += del;
                    if (abs(del) < abs(ans)*EPS) return ans;
                }
                throw("series failed in expint");
            }
        }
    }
    return ans;
}
```

A good algorithm for evaluating Ei is to use the power series for small x and the asymptotic series for large x . The power series is

$$Ei(x) = \gamma + \ln x + \frac{x}{1 \cdot 1!} + \frac{x^2}{2 \cdot 2!} + \cdots \quad (6.3.10)$$

where γ is Euler's constant. The asymptotic expansion is

$$Ei(x) \sim \frac{e^x}{x} \left(1 + \frac{1!}{x} + \frac{2!}{x^2} + \cdots \right) \quad (6.3.11)$$

The lower limit for the use of the asymptotic expansion is approximately $|\ln \text{EPS}|$, where EPS is the required relative error.

```
Doub ei(const Doub x) { expint.h
Computes the exponential integral Ei(x) for x > 0.
    static const Int MAXIT=100;
    static const Doub EULER=0.577215664901533,
        EPS=numeric_limits<Doub>::epsilon(),
        FPMIN=numeric_limits<Doub>::min()/EPS;
    Here MAXIT is the maximum number of iterations allowed; EULER is Euler's constant  $\gamma$ ; EPS
    is the relative error, or absolute error near the zero of Ei at  $x = 0.3725$ ; FPMIN is a number
    close to the smallest representable floating-point number.
    Int k;
    Doub fact,prev,sum,term;
    if (x <= 0.0) throw("Bad argument in ei");
    if (x < FPMIN) return log(x)+EULER;
    if (x <= -log(EPS)) {
        sum=0.0;
        fact=1.0;
        for (k=1;k<=MAXIT;k++) {
            fact *= x/k;
            term=fact/k;
            sum += term;
            if (term < EPS*sum) break;
        }
        if (k > MAXIT) throw("Series failed in ei");
        return sum+log(x)+EULER;
    } else {
        sum=0.0;
        term=1.0;
        for (k=1;k<=MAXIT;k++) {
            prev=term;
            term *= k/x;
            if (term < EPS) break;
            Since final sum is greater than one, term itself approximates the relative error.
            if (term < prev) sum += term;
            else {
                sum -= prev;
                break;
            }
        }
        return exp(x)*(1.0+sum)/x;
    }
}
```

Special case: Avoid failure of convergence test because of underflow.
Use power series.

Use asymptotic series.
Start with second term.

Still converging: Add new term.

Diverging: Subtract previous term and exit.

CITED REFERENCES AND FURTHER READING:

Stegun, I.A., and Zucker, R. 1974, "Automatic Computing Methods for Special Functions. II. The Exponential Integral $E_n(x)$," *Journal of Research of the National Bureau of Standards*,